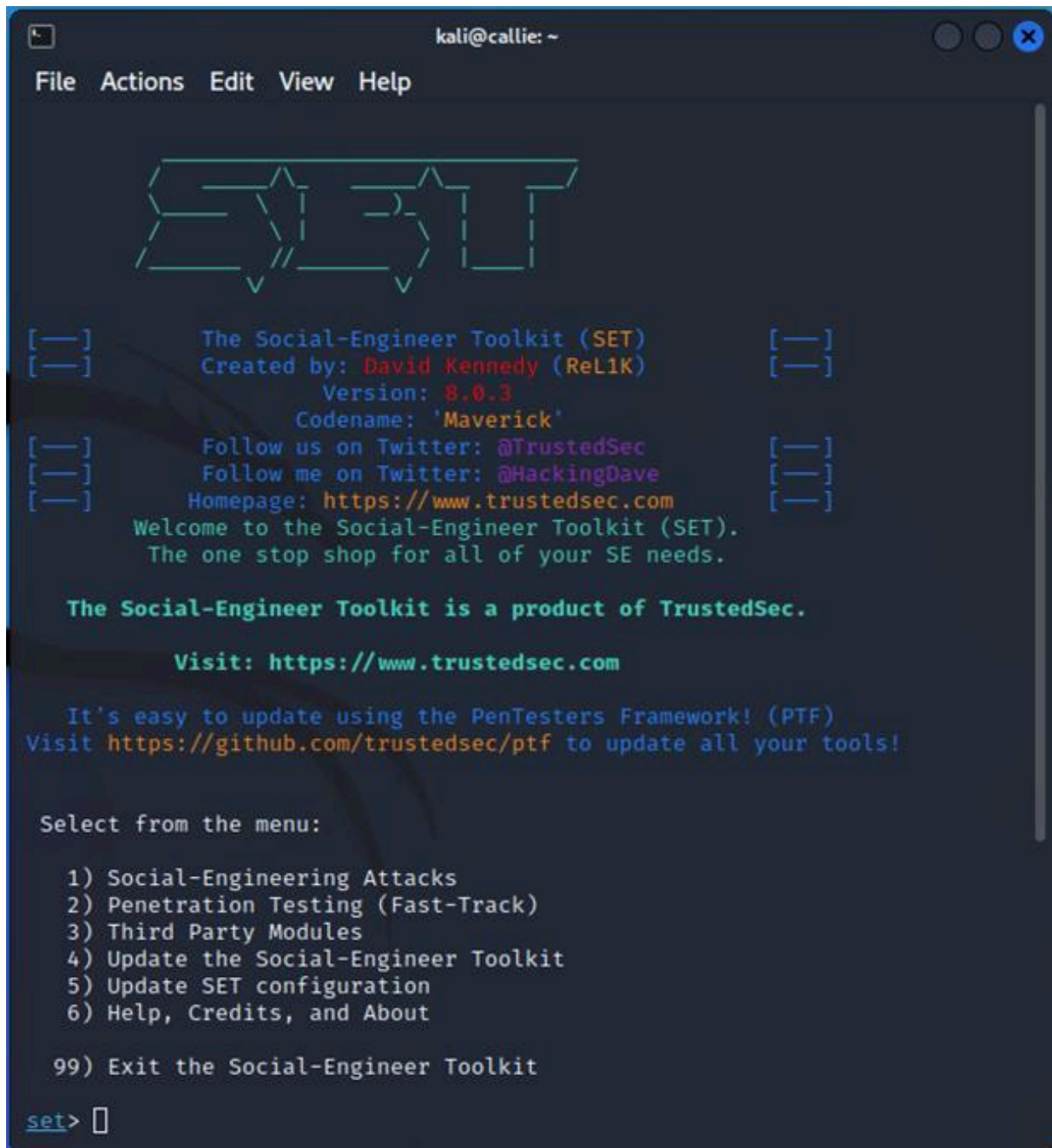


Social Engineering Toolkit (SET)

The **Social Engineering Toolkit (SET)** is an open-source penetration testing framework designed to simulate **social engineering attacks** in **authorized security assessments and training**. It was created by David Kennedy.

A screenshot of the Social Engineering Toolkit (SET) terminal interface. The window title is 'kali@callie: ~'. The menu bar includes 'File', 'Actions', 'Edit', 'View', and 'Help'. The main content area features a large, stylized 'SET' logo at the top. Below the logo, there is a series of lines of text providing information about the toolkit, including its creator (David Kennedy), version (8.0.3), codename ('Maverick'), and social media links. A welcome message follows, stating 'Welcome to the Social-Engineer Toolkit (SET). The one stop shop for all of your SE needs.' Below this, it says 'The Social-Engineer Toolkit is a product of TrustedSec.' and provides a visit link to 'https://www.trustedsec.com'. Further down, it mentions 'It's easy to update using the PenTesters Framework! (PTF)' and provides a link to 'https://github.com/trustedsec/ptf'. At the bottom, there is a 'Select from the menu:' prompt followed by a numbered list of options: 1) Social-Engineering Attacks, 2) Penetration Testing (Fast-Track), 3) Third Party Modules, 4) Update the Social-Engineer Toolkit, 5) Update SET configuration, 6) Help, Credits, and About, and 99) Exit the Social-Engineer Toolkit. The prompt 'set>' is visible at the bottom left.

```
kali@callie: ~
File Actions Edit View Help

  SET

[—] The Social-Engineer Toolkit (SET) [—]
[—] Created by: David Kennedy (ReL1K) [—]
      Version: 8.0.3
      Codename: 'Maverick'
[—] Follow us on Twitter: @TrustedSec [—]
[—] Follow me on Twitter: @HackingDave [—]
[—] Homepage: https://www.trustedsec.com [—]
Welcome to the Social-Engineer Toolkit (SET).
The one stop shop for all of your SE needs.

The Social-Engineer Toolkit is a product of TrustedSec.

Visit: https://www.trustedsec.com

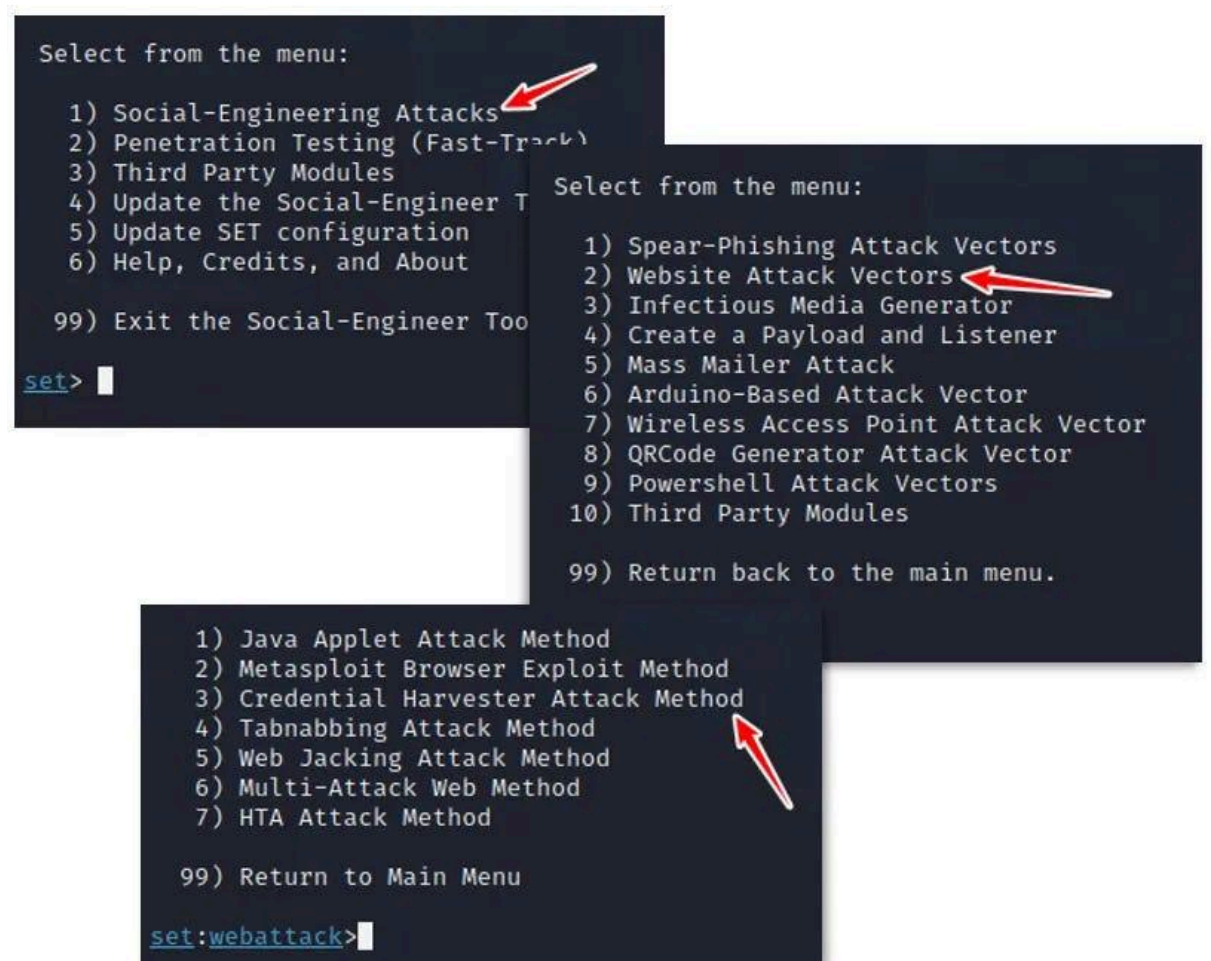
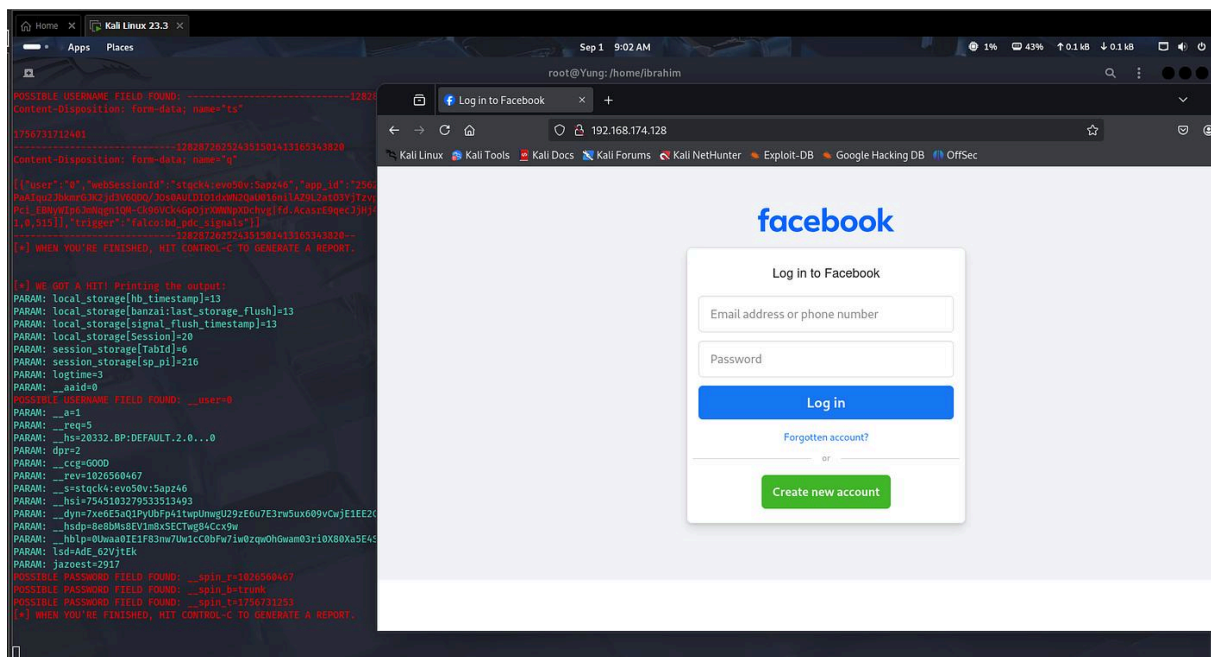
It's easy to update using the PenTesters Framework! (PTF)
Visit https://github.com/trustedsec/ptf to update all your tools!

Select from the menu:

1) Social-Engineering Attacks
2) Penetration Testing (Fast-Track)
3) Third Party Modules
4) Update the Social-Engineer Toolkit
5) Update SET configuration
6) Help, Credits, and About

99) Exit the Social-Engineer Toolkit

set> 
```



What is SET?

SET automates many social engineering scenarios that security professionals use to test how vulnerable an organization is to phishing, malicious USB devices, and other human-targeted attacks.

Common use cases include:

- Phishing awareness assessments
- Security awareness training
- Red team engagements
- Internal penetration testing
- Demonstrating social engineering risks

It is **not** intended for attacking systems without authorization.

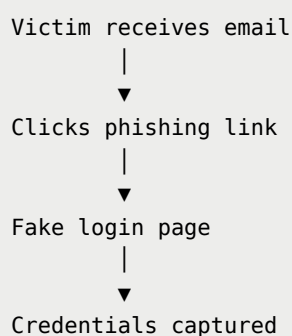
Main Features

1. Spear-Phishing Attack Vectors

SET can help simulate phishing campaigns by:

- Creating realistic login pages
- Sending test emails (where authorized)
- Tracking user interaction
- Collecting submitted credentials during approved assessments

Typical workflow:



Purpose:

- Measure employee awareness

- Improve phishing defenses
-

2. Credential Harvester

The Credential Harvester module hosts a cloned login page that records credentials entered by test participants during an authorized engagement.

Example assessment:

```
https://company-login.example  
  
↓  
  
User enters:  
  
Username:  
john.doe  
  
Password:  
*****  
  
↓  
  
Assessment records the submission
```

This demonstrates the effectiveness of phishing awareness programs.

3. Website Cloner

SET can clone publicly accessible web pages for use in phishing simulations.

It typically copies:

- HTML
- CSS
- Images
- JavaScript (where feasible)

Organizations use this to create realistic awareness exercises.

4. USB / HID Attack Generation

SET can generate payloads for programmable USB Human Interface Device (HID) hardware used in security training, such as:

- Rubber Ducky–style devices
- BadUSB-compatible hardware
- Other keyboard-emulating devices

Example uses include demonstrating:

- Automatic keystroke injection
- Endpoint security weaknesses
- USB device risks

5. Payload Generation

SET integrates with frameworks such as Metasploit to generate payloads for **authorized lab environments and penetration tests**.

These payloads are commonly used to validate security controls and detection capabilities.

6. QR Code Generation

SET can generate QR codes pointing to assessment landing pages.

Example awareness scenario:

```
graph TD; A[Scan QR Code] --> B[Training Login Page]; B --> C[User learns how QR phishing works];
```

This is useful for educating users about "quishing" (QR-code phishing).

7. Web Templates

SET includes templates modeled after common services for awareness testing, allowing organizations to simulate recognizable login experiences during internal exercises.

Typical Menu Structure

The menu varies by version, but commonly includes options such as:

1. Spear-Phishing Attacks
2. Website Attack Vectors
3. Infectious Media Generator
4. Create a Payload and Listener
5. Mass Mailer Attack
6. Arduino-Based Attack
7. Wireless Access Point Attack
8. QRCode Generator Attack
9. PowerShell Attack Vectors

Some options may be unavailable or differ depending on the installed version.

Installation on Kali Linux

SET is preinstalled on many Kali Linux releases.

Launch it with:

```
sudo setoolkit
```

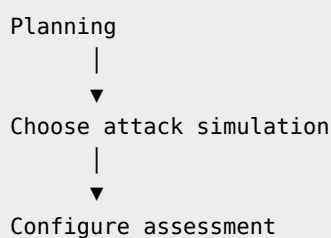
If it is not installed:

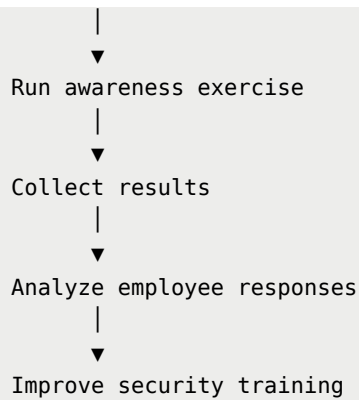
```
sudo apt update  
sudo apt install set
```

Check the version:

```
setoolkit --version
```

Typical Workflow





Advantages

- Easy to use
- Menu-driven interface
- Fast setup for awareness exercises
- Integrates with other penetration testing tools
- Useful for red team and security training

Limitations

- Many email providers have strong anti-phishing protections that can limit testing.
- Modern browsers and endpoint protection may detect or block simulated phishing pages.
- It should only be used with explicit authorization.

Learning Resources

- [TrustedSec – Social Engineering Toolkit](#)
- [Kali Linux Tool Documentation](#)

These resources cover installation, configuration, and legitimate security testing use cases.